

Pramo Jewels

Payment Failure Handling

Document ID: PJ-PFH-001
Version: 1.0
Status: Approved for Development

Field	Value
Project	Pramo Jewels Premium E-Commerce Platform
Purpose	Define procedures for detecting, handling and recovering from payment failures
Audience	Backend Developers, QA, DevOps, Finance, Customer Support
Version	1.0

Failure Scenarios & Responses

Failure Scenario	System Response
Gateway Timeout	Mark transaction pending and retry reconciliation.
Payment Declined	Allow customer to retry or choose another payment method.
Network Failure	Retry safely using idempotency keys.
Webhook Delay	Keep order pending until verified.
Duplicate Callback	Ignore duplicates using event IDs/idempotency.
Refund Failure	Retry automatically and escalate if unresolved.
Settlement Mismatch	Flag for finance reconciliation and investigation.

Objectives

Minimize customer impact, prevent duplicate payments, maintain financial accuracy and ensure reliable order processing.

Recovery Workflow

Detect Failure → Log Event → Classify Error → Retry (where appropriate) → Reconcile with Gateway → Notify Customer → Close Incident.

Retry Strategy

Retry only transient failures using exponential backoff. Never repeat completed transactions; protect all retries with idempotency keys.

Customer Experience

Display clear, non-technical error messages, preserve cart contents and provide retry or alternate payment options.

Security

Validate gateway responses, verify webhook signatures, encrypt sensitive data and protect payment credentials using secure secret management.

Monitoring

Track payment failure rate, retry success rate, timeout frequency, reconciliation delays and gateway availability.

Integrations

Integrates with Payment Gateway, Orders, Cart & Checkout, Notifications, Accounting, Analytics and Fraud Monitoring.

Acceptance Criteria

Failed payments never create duplicate orders, successful retries synchronize correctly, reconciliation resolves pending transactions and all failures are audit logged.